Databases - Exam 2 July 2026 - 90 minutes - Total 31 points

Q1 (1 pt). Which normal form forbids partial dependencies? a) 1NF b) 2NF c) 3NF

Q2 (1 pt). A foreign key must reference: a) a primary or unique key b) any column

Schema: Student(sid, name), Course(cid, title, teacher), Enrollment(sid, cid). Q3 (4 pts). Relational algebra: list the names of students enrolled in every course taught by Rossi.

Schema: Employee(eid, name, dept, salary, hire_date). Q4 (10 pts). SQL: for each department, the average salary of employees hired after 2020, only departments with at least 5 such employees, ordered by average.

Q5 (7 pts). Draw the ER diagram for a library: books, copies, members, loans with dates.

Schema: Member(mid, name), Loan(lid, mid, copy_id, loan_date, return_date). A loan is active while return_date IS NULL. Q6 (4 pts). Write a trigger that prevents a member from having more than 3 active loans.